

a

Semblance 2.0

Drug recommendations from GDSC2 + CTRP single-agent screening × CCLE tumor-cell profiles.

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 **Cell Line Lookup** — pick a profiled cancer cell line and get its ranked drugs: **broad-spectrum** (what its cancer type is generically most sensitive to) plus **mechanism-specific** (drugs with a validated expression signature in that cancer type). ·  **Signature Match** — paste a GSEA program and get the drugs whose drug-sensitive-cell signature it most closely resembles, by biomedical semantic similarity.

 Research / education only — **NOT a clinical tool** and NOT medical advice. Sign convention: **lower / negative = more sensitive**.

 Cell Line Lookup  Signature Match

Pick a profiled cell line — get its ranked drug recommendations (broad-spectrum + mechanism).

Cell line

A375 · Melanoma (SKCM) ▾

Precomputed and served straight from the CDN — no backend, always on.

A375 · Melanoma (SKCM) · 648 drugs tested on this line · 53 cell lines of this type

Broad-spectrum — drugs this cancer type is most sensitive to (cancer-type average)

#	DRUG	TARGET	TYPE AVG	N	THIS LINE	IC50 μM
1	Dabrafenib	BRAF	-2.24	62	-5.96	0.0315
2	SB590885	BRAF	-2.19	33	-3.71	1.9185
3	PD0325901	MEK1, MEK2	-1.74	33	-2.65	0.0239
4	ERK_2440	ERK1, ERK2	-1.52	20	-2.71	0.4091
5	ERK_6604	ERK1, ERK2	-1.50	20	-2.53	1.0367
6	THR-103	Mutant RAS	-1.48	21	-0.48	37.5888
7	SCH772984	ERK1, ERK2	-1.43	33	-2.11	0.4043
8	Refametinib	MEK1, MEK2	-1.41	33	-2.46	0.1706
9	Trametinib	MAP2K1, MAP2K2	-1.29	64	-2.67	0.0077
10	Ulixertinib	ERK1, ERK2	-1.23	33	-1.64	1.3569
11	PLX-4032	BRAF	-1.21	45	-0.71	—
12	Selumetinib	MAP2K1, MAP2K2	-1.14	79	-1.37	0.6351
13	MLN2480	ARAF, BRAF, RAF1	-1.07	30	—	—
14	VX-11e	ERK2	-0.98	33	-1.95	1.3903
15	LJI308	RSK2, RSK1, RSK3	-0.84	33	-0.44	68.5658

Type avg = average sensitivity across all lines of this cancer type (the recommendation), over **n** lines; **This line** = this cell line's own value. Both standardized — **more negative = more sensitive**; IC50 (μM): lower = more potent. Ranks by average potency, so broadly cytotoxic agents can appear high without being lineage-specific — weigh the reported target and n. (Non-therapeutic lab-tool compounds are filtered out.)

Mechanism-specific — drugs with a validated expression signature in this cancer type (exploratory)

DRUG	TARGET	SIGNAL (p)	EVIDENCE (z)
ML162	—	0.71	4.1
1S,3R-RSL-3	GPX4	0.70	3.9
Selumetinib	MAP2K1, MAP2K2	0.64	3.6
PD318088	MAP2K1, MAP2K2	0.56	3.6
GDC-0879	BRAF	0.58	3.6
ML210	—	0.63	3.5
Trametinib	MAP2K1, MAP2K2	0.57	3.3
BI-2536	PLK1	0.49	3.3
PLX-4720	BRAF	0.50	3.2
tacedinaline	HDAC1, HDAC2, HDAC3	0.52	3.2
Merck60	HDAC1, HDAC2	0.50	3.1
topotecan	TOP1	0.44	3.0
SGX-523	MET	0.54	3.0
triazolothiadiazine	PDE4A, PDE4B, PDE4D	0.52	2.9
TG-101348	JAK2	0.48	2.9
CHIR-99021	GSK3B	0.52	2.9
PLX-4032	BRAF	0.42	2.8
Telomerase Inhibitor IX	TERT	0.44	2.7
UNC0638	EHMT1, EHMT2	0.45	2.7
Docetaxel	Microtubule stabiliser	0.48	2.7

All shown clear a permutation test at **p ≤ 0.01**. **Signal (p)** = how well the cancer-type expression signature tracks sensitivity; **Evidence (z)** = standard deviations above a tissue-matched null (higher = stronger, ranks the list). Hypothesis-generating, not a per-patient prediction — and not corrected for testing thousands of drug×cancer pairs.

b

 Cell Line Lookup  Signature Match

Bring a GSEA program — get drugs whose sensitive-cell signature it most resembles (mechanism match, exploratory).

Paste a GSEA results table — clusterProfiler / fgsea schema (ID, Description, NES, pvalue, p.adjust...), CSV or TSV

```
ID,Description,NES,pvalue,p.adjust
REACTOME_MITOCHONDRIAL_TRNA_AMINOACYLATION,REACTOME_MITOCHONDRIAL_TRNA_AMINOACYLATION,2.113,0.001,0.01
REACTOME_TRNA_PROCESSING,REACTOME_TRNA_PROCESSING,2.085,0.001,0.01
REACTOME_MITOCHONDRIAL_TRANSLATION,REACTOME_MITOCHONDRIAL_TRANSLATION,2.074,0.001,0.01
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT,REACTOME_RESPIRATORY_ELECTRON_TRANSPORT,2.064,0.001,0.01
REACTOME_RESPONSE_OF_EIF2AK4_GCN2_TO_AMINO_ACID_DEFICIENCY,REACTOME_RESPONSE_OF_EIF2AK4_GCN2_TO_AMINO_ACID_DEFICIENCY,-2.06,0.001,0.01
REACTOME_ACTIVATION_OF_ATR_IN_RESPONSE_TO_REPLICATION_STRESS,REACTOME_ACTIVATION_OF_ATR_IN_RESPONSE_TO_REPLICATION_STRESS,2.059,0.001,0.01
REACTOME_TRNA_AMINOACYLATION,REACTOME_TRNA_AMINOACYLATION,2.052,0.001,0.01
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New here? [↪ Load an example melanoma signature](#), then hit Search.

...or upload a file

Choose FileNo file chosen

Cancer type (optional)

SKCM

Top K

10

Search

35 pathways in your signature

Best drug matches — drugs whose sensitivity signature most resembles yours

#	DRUG	CANCER TYPE	MATCH	SHARED PATHWAYS
1	PD318088	Melanoma (SKCM)	0.988	Mitochondrial TranslationRespiratory Electron TransportActivation Of The Pre Replicative Complex
2	GDC-0879	Melanoma (SKCM)	0.988	Respiratory Electron TransportMitochondrial TranslationDna Replication
3	Selumetinib	Melanoma (SKCM)	0.982	Dna ReplicationG2 M CheckpointsActivation Of The Pre Replicative Complex
4	Ulixertinib	Melanoma (SKCM)	0.982	Activation Of The Pre Replicative ComplexMitochondrial TranslationActivation Of Atr In Response To Replication Stress
5	etomoxir	Melanoma (SKCM)	0.981	Mitochondrial TranslationRespiratory Electron TransportTrna Aminoacylation
6	Dabrafenib	Melanoma (SKCM)	0.979	Activation Of The Pre Replicative ComplexActivation Of Atr In Response To Replication StressDna Replication
7	MLN2480	Melanoma (SKCM)	0.979	Respiratory Electron TransportAerobic Respiration And Respiratory Electron TransportSialic Acid Metabolism
8	purmorphamine	Melanoma (SKCM)	0.979	Mitochondrial TranslationRespiratory Electron TransportDefects In Vitamin And Cofactor Metabolism
9	istradefylline	Melanoma (SKCM)	0.978	Mitochondrial TranslationRespiratory Electron TransportTrna Processing
10	EX-527	Melanoma (SKCM)	0.976	Respiratory Electron TransportMitochondrial TranslationTrna Aminoacylation

 Exploratory / hypothesis-generating: semantic similarity of your up-program to drug-sensitive-cell signatures. Not a validated predictor and NOT clinical advice.